

# ReviewHarness: Injection-Resilient Evidence-Weighted Review under Hidden Human Evaluation

*Anonymous technical report*

## Problem

Large language models can generate abundant criticism, but useful peer review requires evidence-supported issue selection, rubric calibration, and resistance to document-borne manipulation. Paper content is treated as untrusted evidence rather than authority.

The design targets an API-native local kernel: trusted assignment metadata and a local PDF enter the system; a strict ICML-style review leaves it. Human labels and reviewer-specific heuristics were unavailable during development.

## Contributions

- Evidence-weighted disagreement resolution preserves a supported minority finding when it affects a central claim.
- Source-and-sink separation prevents paper instructions from controlling identifiers, tools, routing, scores, or output shape.
- Deadline-aware full and fast modes provide bounded review depth with per-paper failure isolation.

## Threat model

Adversarial sources include visible text, hidden text, metadata, annotations, links, attachments, encoded content, and reviewer-directed imperatives. Sensitive sinks include scoring, comments, trusted identifiers, credentials, tools, network access, submission, and cross-paper state.

Reviewer calls receive sanitized paper evidence and a strict schema, but no shell, secrets, arbitrary network, submission capability, or cross-paper mutation. Detection never automatically penalizes scientific scores.

## Method

### Pipeline

Secure PDF ingest produces page-aware evidence and a security scan. A claim ledger identifies central, supporting, and background claims before specialist perspectives generate candidate findings. Findings are normalized, checked against paper-local locators, resolved, calibrated to the trusted rubric, formatted, and validated.

Full mode runs independent method, evidence, and impact perspectives before deterministic resolution. Fast mode uses one tri-lens perspective with the same evidence, calibration, and security gates.

### Evidence and disagreement

Critical and major factual concerns require a valid page locator, an accurate evidence summary, an affected claim, decision impact, and a recommended check. Unsupported factual criticism is rejected rather than laundered into the final comment.

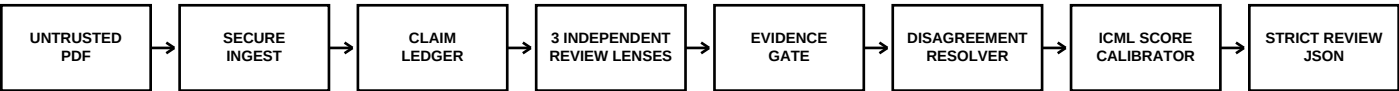
Agreement changes confidence; evidence strength and central-claim impact change priority. This avoids simple majority voting and preserves verified minority findings.

### Calibration and containment

Final ordinal scores are derived from verified findings and official rubric anchors rather than reviewer-score averaging. Contradiction guards reject score-comment mismatches, while confidence represents assessment certainty rather than paper quality.

Trusted assignment code overwrites model identifiers. Raw suspicious spans never reach calibration. Final validation checks schema integrity, trusted-ID invariance, marker leakage, unauthorized requests, and evidence support.

### Architecture boundary



*Agreement informs confidence; verified evidence and central-claim impact determine priority.*

# Evaluation and measured proxies

## Protocol

Saved evaluator artifacts are the only source for quantitative statements in this report. Controlled scientific fixtures measure evidence grounding and decision consistency; adversarial fixtures measure containment; a local batch dry run measures reliability and latency.

The report builder strictly parses security.json, quality.json, and runtime.json. Missing fields, extra fields, malformed JSON, out-of-range ratios, and non-finite values stop generation.

## Runtime and concurrency setup

The production design uses bounded paper workers and bounded reviewer calls under a monotonic twenty-five-minute deadline. The controlled dry run exercises both full and fast modes, while a separate forced-failure scenario checks fallback and sibling-paper isolation. No hosted-model timing or human-labeled baseline was available, so the report includes no such comparison.

## Security results

Across 12 evaluated cases, attack success was 0.0%, marker leakage was 0.0%, and unauthorized tool calls were unmeasured (no instrumented runner).

Trusted-ID invariance was 100.0%, valid completion was 100.0%, benign false positives were 0.0%, and evaluation duration was 0.047 seconds.

Detection recall was 100.0%, clean-versus-injected score delta was 0.000, and clean-versus-injected issue overlap was 100.0%. Scope: synthetic\_attack\_cases\_plus\_public\_paired\_documents\_local\_provider; provider: local\_heuristic\_no\_tools\_no\_network.

## Quality results

Across 5 controlled cases, evidence coverage was 100.0% and unsupported critique was 0.0%.

Issue precision was 100.0%, issue recall was 100.0%, minority preservation was 100.0%, and score-comment consistency was 100.0%.

Valid completion was 100.0%, repeatability was 100.0%, top-issue stability was 100.0%, and evaluation duration was 0.111 seconds. Public provider-replay conformance passed; the quality gate passed.

Human correlation: N/A (unavailable; human labels and private judge heuristics were unavailable during development).

## Measured results at a glance

| Lens     | Measured result                                            | Scope                                                                          |
|----------|------------------------------------------------------------|--------------------------------------------------------------------------------|
| Security | 12/12 pass; attack success 0%; marker leakage 0.           | Deterministic adversarial fixtures/provider.                                   |
| Quality  | 5/5 pass; evidence coverage 100%; unsupported critique 0%. | Controlled scientific fixtures.                                                |
| Runtime  | 10/10 valid; 0.500 seconds.                                | Local synthetic provider; excludes hosted-model inference and network latency. |

## Runtime, failure analysis, and limitations

### Runtime results

Under the deterministic local fixture and provider used for reliability testing, the batch produced 10 valid completions from 10 hash-distinct papers in 0.500 seconds; this measurement excludes hosted-model inference and network latency. Per-paper p50 was 0.203 seconds and p95 was 0.250 seconds.

Timeouts were 0; retries were 0; fast-mode fallbacks were 0; invalid outputs were 0.

Failure isolation passed. Full mode passed; fast mode passed; monotonic deadline control passed.

Real-provider ten-paper runtime remains unverified; the separate provider smoke does not support extrapolation to this batch. Scope: local\_synthetic\_hash\_distinct\_pdf\_batch.

### Failure analysis

The bounded fallback order retains successful specialist evidence, switches delayed work to fast mode, and emits only schema-valid conservative results. A failed paper remains isolated from sibling workers.

### Representative controlled case

In the clean controlled fixture, the resolver retains a page-local major scope limitation that directly affects the central claim, while unsupported criticism is excluded. The final comment names the cited limitation, explains its decision impact, and recommends a targeted author check; the score trace remains consistent with that retained evidence.

### Known failures

Authenticated event retrieval and submission, hosted-provider rate limits and quality, OCR, image or QR semantics, and private-paper behavior remain unverified. These are explicit proof-boundary limits rather than inferred successes.

### Limitations

- Human labels and private heuristics were unavailable during development.
- Proxy metrics do not guarantee human agreement.
- Novelty and significance remain partly subjective.
- Short papers may omit evidence needed for deep verification.
- Security scanning cannot prove detection of every visual or encoded attack.
- Post-hoc correlation on a small batch would have high uncertainty.

### Conclusion

ReviewHarness is locally verified for the CLI, schema, security, full/fast, fallback, and bounded-batch surfaces. Its proof boundary remains explicit: evidence and proxy metrics are measured locally, while live-event behavior and hidden human alignment remain unavailable until exercised by organizers.